

SATTWIK BISHNU

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📍 Kolkata, India

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SUMMARY

Electronics and Communication Engineering student with strong foundation in Data Structures, Algorithms, Object-Oriented Programming, and Embedded Systems. Experienced in developing backend applications using Java, Python, and MySQL along with Arduino-based IoT systems. Seeking software engineering, backend development, or embedded systems internship opportunities where I can apply programming, problem-solving, and embedded system design skills.

EDUCATION

Institute of Engineering and Management, Kolkata

2023-2027

- B.Tech in Electronics and Communication Engineering. CGPA- 8.81

East West Model School

- ISC (Class XII) – 78.8% | 2023
- ICSE (Class X) – 87.8% | 2021

TECHNICAL SKILLS

Programming Languages: C, C++, Java, Python, MATLAB, JavaScript

Web Development: HTML, CSS, JavaScript, Flask

Databases: MySQL

Libraries & Tools: NumPy, Pandas, Git, GitHub, VS Code, PyCharm, IntelliJ IDEA

Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems

Hardware & Embedded Systems: Arduino, IoT Prototyping, Sensor Integration

TECHNICAL PROJECTS

Smart Library Management System:

- **Technologies:** Python, MySQL
- Developed a library management system for managing book inventory, user records, and transaction history.
- Implemented automated book issue and return tracking with database integration.
- Designed a recommendation feature to suggest books based on user borrowing patterns

Bank Management System

- **Technologies:** Java, MySQL
- Developed a Bank Management System for managing customer accounts and banking operations.
- Implemented CRUD operations, JDBC connectivity, and secure authentication mechanisms.
- Designed a modular system using Object-Oriented Programming principles to handle 500+ simulated transactions efficiently

Amazon Shopping Cart Prototype

- **Technologies:** HTML, CSS, JavaScript
- Built a dynamic shopping cart interface allowing product addition/removal, quantity updates, and bill generation.

AI-Based Trash Classifier

- **Technologies:** Arduino, Python, TensorFlow, OpenCV.
- Developed a waste classification system using webcam-based detection and servo motor automation to segregate materials.

CERTIFICATIONS

- Python Programming & Data Structures – University of Michigan (2025).
- Information Theory – The Chinese University of Hong Kong (2025).
- Advanced System Security Topics – University of Colorado (2024).
- Classical Cryptosystems and Core Concepts – University of Colorado (2025).

MEMBERSHIPS & LEADERSHIP

- IEEE Electromagnetic Compatibility Society (EMC-SBC) - Treasurer.
- IEEE Antennas and Propagation Society (APS) - Member